W-F-1234A
October 2, 1973
SUPERSEDING
Int. Fed. Spec. W-F-001234
February 11, 1967 and
Fed. Spec. W-L-331
July 11, 1950

FEDERAL SPECIFICATION

FIXTURE, LIGHTING (FLUORESCENT LAMP, INDUSTRIAL)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers industrial lighting fixtures of the fluorescent lamp type for alternating current service, as specified (see 3.3.11). Unless otherwise specified, lamps shall not be included with the fixtures.

1.2 Classification. Fixtures furnished under this specification shall be suspension mounted and of the following types, styles, classes, kinds, and sizes as specified (see 6.1):

Types:

I - Individual mounting
II - Continuous row mounting

Styles:

A - Closed-end reflector
B - Open-end reflector

Classes:

I - Porcelain enamel reflector finish
II - Organic enamel reflector finish
III - Aluminum oxide reflector finish

Kind:

Kind A - 8-15 percent of light output in upward direction, 92-85 percent downward.

Size 1 - 2 lamp (40 watt, 48 in., T-12 lamps)
Size 2 - 2 lamp (800 ma., 96 in., T-12 lamps)
Size 3 - 3 lamp (40 watt, 48 in., T-12 lamps)
Size 4 - 3 lamp (800 ma., 96 in., T-12 lamps)
Size 5 - 2 lamp (1.5 ampere, 96 in., T-12 or PG-17 lamps)

Kind B - 18-25 percent of light output in upward direction, 82-75 percent downward.

Size 1 - 2 lamp (40 watt, 48 in., T-12 lamps)
Size 2 - 2 lamp (800 ma., 96 in., T-12 lamps)
Size 5 - 2 lamp (1.5 ampere, 96 in., T-12 or PG-17 lamps)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

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Federal Specifications:

W-B-30 - Ballast, Fluorescent Lamp.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-cleated
PPP-B-601 - Boxes, Wood, Cleated-plywood.
PPP-B-621 - Boxes, Wood, Nail and Lock-corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-wall.
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container grade) and Cut Shapes.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling, and Testing.

Fed. Test Method Std. No. 151 - Metals: Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc., Standards (UL):

UL 57 - Electric Lighting Fixtures.
UL 542 - Lampholders, Starters, and Starter Holders for Fluorescent Lamps.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, L.I., New York 11746, 207 East Ohio Street, Chicago, Illinois 60611; or 1655 Scott Boulevard, Santa Clara, California 95050.)

American National Standards Institute (ANSI) Inc. Standards:

C78.700 - 40-Watt T-12 Rapid-Start Fluorescent Lamp.
C78.702 - 96-Inch (800-Milliampere) T-12 Rapid-Start Fluorescent Lamp.
C78.707 - 96-Inch (1.5-Ampere) T-12 and PG-17 Rapid-Start Fluorescent Lamps.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018)

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Illuminating Engineering Society (IES) Handbook:

(Application for copies should be addressed to the Illuminating Engineering Society, 345 East 47th Street, New York, New York 10017.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification

(Application for copies should be addressed to the National Freight Traffic Association, Inc., Agent, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 Riverside Plaza, Chicago, IL 60606.)

Porcelain Enamel Institute, Inc.

PEI Bulletin T-29(72) - Test For The Adherence of Porcelain Enamel Cover Coats Direct-To-Steel.

(Application for copies should be addressed to Porcelain Enamel Institute, Inc., 1900 L Street, NW., Washington, DC.)

3. REQUIREMENTS

3.1 Fire, Safety and Casualty Hazards. Each contractor shall submit to the contracting agency proof that the fixture he proposes to supply under this specification conforms to the requirements of the Underwriters' Laboratories, Inc., (UL) Standard No. 57 - Electrical Lighting Fixtures. The listing mark of the Underwriters' Laboratories, Inc. affixed to each fixture will be accepted as evidence that the product conforms to this requirement. Compliance with the above preliminary requirements in regard to fire, safety and casualty hazards does not absolve the contractor from complete compliance with the other requirements of this specification in order to secure acceptance of his material or equipment.

3.2 Design, general.

3.2.1 Accessibility of parts. Each lamp shall be easily removable and replaceable without the use of tools and without the removal of other lamps. The ballast(s) and wiring shall be easily accessible without the use of special purpose tools. Additionally, lampholders and ballast(s) shall be installed in a manner that will allow replacement without removal of the fixture from the installed position.

3.2.2 Bolts, nuts, and lockwashers. Bolts shall have either two nuts or one nut and a lockwasher.

3.2.3 Prevention of dust collection. Fixtures shall be of a ventilated type having top openings. Fixtures shall be designed to minimize dirt and dust collection on lamp and reflector surfaces by allowing an air draft path sufficient to move dust and dirt particles upward and through the fixture to the outer air.

3.3 Assembly and construction. Unless otherwise indicated, the fixture shall be furnished with reflector(s), enclosure for ballast(s) and wiring, lampholders, ballast(s), hangers, wiring and all other necessary parts required to constitute a complete functional luminaire. Each fixture shall be completely assembled and ready for installation and connection to building wiring with exception of hardware specifically required for mounting the fixture. The construction of fixtures shall conform to applicable requirements of UL 57. Additionally, housing of fixture shall be provided with a separate factory installed grounding device. The grounding device is to be used for connecting a separate grounding conductor to the fixture housing and for no other purpose. The device shall meet all applicable grounding requirements of the National Electrical Code, latest edition. A complete parts list shall be furnished with each fixture.

3.3.1 Material. The material of which fixture is constructed shall be free from all defects and imperfections which might affect serviceability or appearance.

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3.3.2 Ballast. Ballast(s) shall conform to Federal Specification W-B-30 and shall be designed to operate on one of the following voltage ranges and alternating current frequencies; 110-125 volt 60 Hertz; 250-290 volts, 60 Hertz; or 210-236 volts, 50 Hertz. The particular voltage range and frequency, one of those cited above, shall be specified (see 3.3.11 and 6.1). Additionally ballast(s) shall be of the rapid-start, high power factor type, having approved built-in protective and automatic reset features. The built-in protective and automatic reset features shall be of the thermostatic type class "P", meeting UL requirements. Also, ballast(s) shall be held firmly in contact with the metal of the fixture by at least 4 thread cutting (TC) sheet metal screws or by at least 4-bolts secured in a manner that precludes the bolts from turning. Ballast requirements per fixture size (see 1.2) are as follows:

3.3.2.1 Size 1. The size 1 fixture shall be equipped with a two-lamp 400 to 450 ma ballast and shall accommodate two 48-inch, 40 watt, T-12 bi-pin lamps in accordance with ANSI Standard C78.700.

3.3.2.2 Size 2. The size 2 fixture shall be equipped with a two-lamp 800 to 830 ma ballast and shall accommodate two 96-inch, T12 recessed double contact lamps in accordance with ANSI Standard C78.702.

3.3.3.3 Size 3. The size 3 fixtures shall be equipped with 400 to 450 ma ballasts and shall accommodate three 48-inch, 40 watt, T-12 bi-pin lamps in accordance with ANSI Standard C78.700.

3.3.2.4 Size 4. The size 4 fixtures shall be equipped with 800 to 830 ma ballasts and shall accommodate three 96-inch, T-12 recessed double contact lamps in accordance with ANSI Standard C78.702.

3.3.2.5 Size 5. The size 5 fixture shall be equipped with a two-lamp 1400 to 1700 ma ballast and shall accommodate two 96-inch, T-12 or PG-17 recessed double contact lamps in accordance with ANSI Standard C78.707.

3.3.3 Lampholders. The lampholders shall be of a depressible contact bi-pin or recessed double contact type in accordance with the applicable requirements of UL 542. The lampholder mountings shall be sufficiently rigid to prevent spreading of the lampholders upon installation of the applicable lamp. The longitudinal distance between the inner surfaces of a pair of lampholders shall be sufficient, but not greater than necessary, to insure proper fit and contact of lamps specified in the applicable ANSI Standard; for sizes 1 and 3, ANSI C78.700; for sizes 2 and 4, ANSI C78.702; and for size 5, ANSI C78.707.

3.3.4 Lamp spacing. The lamps of the size 1, 2, and 5 fixtures and the outside lamps of the size 3 and 4 fixtures shall be located equal distances from the center line of the fixture. The center lamp of the size 3 and 4 fixtures shall be equidistant from the two outside lamps and shall have a shielding angle not less than the outer lamps.

3.3.5 Enclosure and reflectors. Fixture shall be fabricated in accordance with U.L Standard 57. Enclosure shall be constructed of either 20 gage (0.036 inch) steel, 22 gage (0.030 inch) steel with stiffening ribs, or 18 gage (0.040 inch) minimum thickness aluminum. Class I and II reflectors shall be constructed of either 20 gage (0.036 inch) steel or 22 guage (0.030 inch) steel with stiffening ribs. Class III reflectors shall be constructed of either 18 gage (0.040 inch) minimum thickness aluminum or 20 gage (0.032 inch) aluminum with stiffening ribs. See 4.5.3.

3.3.6 Mounting. Each fixture shall be provided with chains, brackets, sliding clamp hangers, conduit stems and canopys, suspension rods or pipes, or other mounting hardware suitable for the mounting method and mounting and ceiling heights specified in the invitation for bids (see 6.1). Sliding clamp hangers, as discussed above, shall be of a design that permits attachment at any point along approximately the entire fixture length.

3.3.6.1 Type I. Type I fixture shall be provided with the specified mounting means for installation as an individual unit.

3.3.6.2 Type II. The type II fixture shall be provided with the specified mounting means for the installation of two or more fixtures in a continuous row. Additionally, fixtures shall be designed to insure alignment of adjoining units and preclude light leakage when mounted in continuous rows.

3.3.7 Center shield. Kind B fixtures shall be equipped with a longitudional center shield. The shield shall be designed and constructed so as to prevent warping and distortion of fixture.

3.3.8 Reflector.

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3.3.8.1 Material and finish.

3.3.8.1.1 Class I porcelain enamel on steel. Class I reflectors shall be finished in accordance with either one of the following two methods. In the first method, the reflecting surface shall be covered by a ground coat of porcelain enamel, fused to the metal at or above a red heat. The reflecting surface shall then be coated with one or more coats of white porcelain enamel, separately fired, which shall cover the ground coat. The non-reflecting surface shall be coated with one or more coats of porcelain enamel, separately fired. The second method requires that reflector be of decarbonized enameling steel (0.01% maximum carbon composition), referred to as "Direct-On Steel." In this method, only one coat of white porcelain enamel, fused to the reflector at or above red heat, need be applied to the decarbonized steel reflector. In either method, the total thickness of porcelain enamel, measured as specified in 4.5.2.1.1 shall not exceed 0.025 inch at any point. All porcelain enameled surfaces shall be free from bare spots, tears, star marks, crazing, blisters, black specks, hairlines, chipping, pinholes, and other defects which affect appearance or which may impair serviceability.

3.3.8.1.2 Class II organic enamel on steel. The Class II reflectors shall be treated chemically to provide a suitable rust-inhibiting base for the finish and shall be covered completely with a baked-on priming coat so as to leave no exposed metal. All surfaces of the reflector shall be covered completely and uniformly with one or more coats of enamel having a urea-formaldehyde, melamine-formaldehyde, or acrylic base. The reflecting surfaces shall be white. All enameled surfaces shall be free from tears, star marks, crazing, blisters, black specks, hairlines, chipping, pinholes, and other defects and irregularities.

3.3.8.1.3 Class III, aluminum oxide on aluminum. The Class III reflectors shall be protected by a dense coating of aluminum oxide weighing not less than 5.0 milligrams per square inch (see 4.5.1.1.1) and further protected by a stain resisting sealing treatment.

3.3.8.1.4 Reflector finish tests. Class I, II and III reflectors shall be subjected to tests specified in 4.5.1.1., 4.5.2.1, 4.5.3.1, and 4.5.3.2.

3.3.8.2 Reflector end shields. Style B, open end reflector, fixtures shall not be equipped with end shields. Each Style A, closed-end reflector, fixture shall be provided with the number of end shields indicated below, dependent upon type. Each type II style A fixture shall be designed such that end shields can be attached and detached at either end of fixture. The number of end shields per each style A fixture shall be:

2 end shields for Type I, Individual mounting
1 detachable end shield for Type II, Continuous row mounting

3.3.8.3 Dimensions. Minimum inside width dimensions of any reflector shall be at least 11-3/4 inches. Length dimension shall be as required to accommodate the lamps specified for each size fixture (see 1.2 and 3.3.2), provided that all requirements of this specification are adhered to.

3.3.8.4 Reflectance. When determined in accordance with 4.5.3.1.2, the average reflectance of test samples after exposure shall decrease by no more than 4 percent of the initial average reflectance before exposure. Also, the average reflectance of test samples after exposure shall be at leat 96 percent of average reflectance value given for fixture in independent laboratory photometric test data (see 4.6.5). When tested in accordance with 4.5.2.1.2, fixture average reflectance shall be at least 96 percent of the average reflectance value given in photometric test data (again, see 4.6.5).

3.3.9 Finish (non-reflecting surfaces). Unless otherwise specified (see 6.1), the finish on all non-reflecting exterior surfaces shall be aluminum oxide on aluminum; white, gray or aluminum paint on steel; nickel or cadmium plating on copper alloy. The aluminum oxide coating shall not weigh less than 5.0 milligrams per square inch (see 4.5.1.2.1). Fastening devices shall be nickel, chromium, cadmium or zinc plated. All painted surfaces shall be free from tears, star marks, crazing, blisters, black specks, hairlines, chipping, pinholes and other defects which may impair appearance or serviceability. In general, finish shall be appropriate for use in an industrial environment.

3.3.9.1 Non-reflecting finish tests. Non-reflecting finishes shall be subjected to tests specified in 4.5.1.2, and 4.5.3.2.

3.3.10 Wiring. The wiring of the fixture shall be in accordance with the applicable requirements of UL 57. When specified in the invitation for bids (see 6.1), the type I fixture shall be furnished with 3 conductor, AWG size 16, flexible cord, of specified length, with male parallel slotted plug connector, in accordance with the applicable requirements of UL 57. The flexible cord and plug and connector shall be assembled to fixture ready for use.

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3.3.11 Electrical requirements. Fixtures shall be designed to operate on one of the following voltage ranges and alternating current frequencies, as specified (see 3.3.2 and 6.1): 110-125 volts, 60 Hertz; 250-290 volts, 60 Hertz; or 210-236 volts, 50 Hertz.

3.3.12 Fixture rigidity. The assembled fixture, with the required number of lamps inserted therein, shall be subjected to the test specified in 4.5.2.2. When tested in the manner outlined neither fixture nor lamps shall become damaged or disassembled.

3.3.13 Marking. Unless otherwise specified (see 6.1) fixture shall be marked in accordance with the provisions of UL - 57. Marking shall be discernable when fixture is assembled but shall not be visible from the floor. Additionally, the marking shall include FSC number, contract or order number, date (month and year) of manufacture and model number.

3.3.14 Interchangeability of parts. The parts of each fixture of a particular model number shall be interchangeable with any other fixture of the same model number produced by the same manufacturer.

3.3.15 Photometric requirements.

3.3.15.1 Shielding angle. Fixture shielding angle, as used herein shall be the angle between the horizontal plane containing the fixtures longitudional axis and the line of sight at which the bare light source (fluorescent lamp(s)) within the fixture first become visible. When determined in accordance with paragraph 4.6.1, the minimum shielding angle for kind A fixtures shall be 13 degrees and minimum shielding angle for kind B fixtures shall be 25 degrees.

3.3.15.2 Spacing-to-mounting height ratio. The spacing-to-mounting height (S/MH) ratio is the ratio of the distance between luminaire centers to the mounting height above the work plane, reference page 1-18 of IES Lighting Handbook, fifth edition 1972. When determined in accordance with paragraph 4.6.2, the S/MH ratio of size 1, 2, 3, and 4 fixtures shall be within the range of 1 to 1.6 and the S/MH ratio of the size 5 fixture shall be within the range 1.2 to 1.6.

3.3.15.3 Light distribution. When determined in accordance with paragraph 4.6.3, the percent of light distribution, expressed as a percentage of the lighting fixture's total lumen output in lumens shall be as follows:

Kind A Fixture: 8-15 percent upward, 92-85 percent downward

Kind B Fixture: 18-25 percent upward, 82-75 percent downward

3.3.15.4 Efficiency. Efficiency as used herein is the ratio, in percent, of luminous flux leaving a fixture to the total luminous flux emitted by the lamp or lamps used therein. When determined in accordance with paragraph 4.6.4 fixtures shall have the minimum efficiencies indicated in Table I.

TABLE I.

Class	Kind	Size	Min. Efficiency
I, II	A	1	86
I, II	A	2	82
I, II	A	3	84
I, II	A	4	78
I, II	A	5	82
I, II	B	1	87
I, II	B	2	84
I, II	B	5	83
III	A	1	72
III	A	2	72
III	A	3	72
III	A	4	72
III	A	5	70
III	B	1	82
III	B	2	82
III	B	5	70

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3.3.16 Preproduction Sample. Unless otherwise specified (see 6.1), before production is commenced, a sample of each type, style, class, kind, and size of fixture specified in the invitation for bids shall be made available to the contracting officer or his authorized representative for approval in accordance with paragraphs 4.4, 4.5, and 4.6. Test reports indicating that all required tests specified or described in paragraphs 4.5 and 4.6 have been successfully completed must accompany each preproduction sample at the time it is made available to the Government. Futhermore, test reports must be obtained through an independent Laboratory acceptable to the Government. The requirements of this paragraph shall be applicable on a per contract basis.

3.3.17 Workmanship. Finished fixture shall be clean and conform to the quality and grade of the product established by this specification. All burrs and rough and sharp edges shall be removed. Exposed welds or seams shall be ground smooth. There shall be no flaking, peeling, or chipping of paint or exposure of bare metal. All holes shall be accurately and cleaning punched or drilled. Fixture(s) shall be free from any defects that affect appearance or serviceability and any defects that may cause injury.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified (see 6.1), in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.

4.2.1 Preproduction sample inspection. The preproduction sample (see 3.3.16) shall be examined for approval in accordance with 4.4, 4.5, and 4.6. Each test cited in 4.5 shall be accomplished on only one fixture and, except as noted in 4.5.3.1.2, on only one test panel representative of the fixture. All tests cited in 4.6 shall be accomplished on one complete fixture. Testing in accordance with 4.5 and 4.6 shall be performed by an independent test laboratory or laboratories acceptable to the Government.

4.2.2 Lot acceptance inspection. Lot acceptance inspection shall consist of a visual examination (4.4), in-process testing in accordance with 4.5.1 and end item testing in accordance with 4.5.2. Samples shall be selected in accordance with 4.3.

4.3 Sampling and acceptance criteria for lot acceptance inspection.

4.3.1 Sampling and acceptance criteria for visual examination. Sampling and acceptance criteria for visual examination (4.4) shall be in accordance with provisions set forth in MIL-STD-105. The Inspection Level shall be level II and the Acceptable Quality Level (AQL) shall be 2.5 percent for major defects and 6.5 percent for minor defects, expressed in terms of percent defective.

4.3.2 Sampling and acceptance criteria for tests. Except as indicated in 4.5.2.2, sampling and acceptance criteria for tests shall be in accordance with MIL-STD-105. The inspection level for in-process testing shall be level S-2 with an AQL of 6.5 percent defective. The inspection level for end item testing shall be level II with an AQL of 2.5 percent defective.

4.3.3 Definition of lot. Lot as used herein shall be the collection of units of products from which samples are to be drawn and inspected to determine compliance with specification requirements. The lot shall consist of units of product of a single Type, Style, Class, Kind and Size (see 1.2). Additionally, sample fixtures shall be from the same production run and test panels (4.5.1.1.2) shall be representative of this run.

4.4 Visual examination. The completely assembled end item shall be examined for, but not limited to the defects listed in table II. The inspection level shall be as specified in 4.3.1.

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4.4.1 Table II, Classification of Defects.

TABLE II.

Defects	Requirement Paragraph	Classification Major	Minor
Parts not accessible	3.2.1	X	
Bolts, lockwashers, nuts not as specified	3.2.2	X	
Not designed to minimize dirt and dust collection	3.2.3	X	
Parts missing	3.3, 3.3.8.2	X	
Not assembled as specified	3.3	X	
Grounding device not as specified	3.3	X	
Material not as specified	3.3.1	X	
Ballast not as specified	3.3.2	X	
Lampholders not as specified	3.3.3	X	
Lamp spacing not as specified	3.3.4	X	
Enclosure and reflector not as specified	3.3.5	X	
Mounting not as specified	3.3.6	X	
Center shield not as specified	3.3.7	X	
Material and finish not as specified	3.3.8.1	X	
Dimensions not as specified	3.3.8.3	X	
Finish on non-reflecting surfaces not as specified	3.3.9		X
Wiring not as specified	3.3.10	X	
Electrical characteristics not as specified	3.3.11	X	
Marking not as specified	3.3.13		X
Parts not interchangeable	3.3.14	X	
Workmanship not as specified	3.3.17	X	

4.5 Tests.

4.5.1 In-process testing.

4.5.1.1 Reflector finish.

4.5.1.1.1 Weight of finish (applicable to Class III fixtures only). The weight per square inch of Class III, aluminum oxide coating shall be determined by comparing the weight of a sample reflector before and after a uniform application of coating and dividing the weight increase by the area of coating.

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4.5.1.1.2 Selection and Preparation of test panels (applicable to class II and III fixtures only). Class II and III test panels of the same material and thickness as specified in 3.3.5, and having a finish in accordance with that specified for the appropriate Class II or III reflector (see 3.3.8.1.2 and 3.3.8.1.3), shall be prepared in accordance with Method 2011 of Fed. Test Method Std. No. 141. Following preparation, Class II panels shall be subjected to tests specified in 4.5.1.1.2.1, 4.5.1.1.2.2 and 4.5.1.1.2.3 and Class III panels shall be subjected to the test specified in 4.5.1.1.2.1. Note: Prior to testing, Class II test panel finish thickness and reflector finish thickness shall be determined in accordance with Method 6181 of Fed. Test Method Std. No. 141, and compared to insure that panel finish thickness is representative of reflector finish thickness.

4.5.1.1.2.1 Impact (applicable to Class II and III fixtures only). Test panel shall be placed over a hole 1-3/4 inches in diameter in a steel anvil, whereupon a steel ball weighing one pound shall be dropped from a height of 16 inches such that it will strike the panel at the center of the hole. The surface of the panel opposite the point of impact shall show no peeling of the finish but may show cracks. Surface peeling shall be construed as test failure.

4.5.1.1.2.2 Adhesion (applicable to Class II fixtures only). On a panel not previously subjected to testing, a 1-inch square lattice made of 1/8 inch squares shall be scratched through the finish to the base metal, using a sharp razor blade. After inscribing the lattice, a sharp rounded knife blade shall be scrapped in a direction diagonal to the scratches whereupon no sections of the finish, within the lattice, shall flake off. Evidence of flaking shall be construed as test failure.

4.5.1.1.2.3 Hardness (applicable to Class II fixtures only). Test panel shall be subjected to test as follows. A drafting pencil equal in hardness to a No. 2H pencil shall be sharpened by trimming away the wood so as to have about 1/4 inch of the lead exposed and undisturbed. Care shall be exercised to avoid scratching or nicking the lead. The end of the lead shall be made smooth, flat, and circular by holding the pencil perpindicular to and rubbing the end on fine sandpaper held on a hard surface. The test shall be performed by holding the pencil in the normal manner for writing and at an angle of 45 degrees to the surface and pushing the pencil against the surface of the panel. The pressure applied can be as desired and will not affect the results. The lead of the pencil shall crush and shall not cut, scratch, or scrape into the surface of a finish of sufficient hardness. Should finish become cut or scratched, it shall be considered to be too soft and shall be construed as failing to meet the requirements of this specification.

4.5.1.2 Non-reflecting finishes.

4.5.1.2.1 Weight of finish (applicable to aluminum oxide finishes only). Weight per square inch of aluminum oxide coating shall be determined by comparing fixture weight (without reflector) before and after application of a uniform coating and dividing weight increase by area of coating.

4.5.1.2.2 Preparation of sample fixtures. Sample fixtures shall be prepared in accordance with method 2011 of Fed. Test Method Std. No. 141. Following preparation, fixtures shall be tested as specified in 4.5.1.2.2.1, 4.5.1.2.2.2 and 4.5.1.2.2.3.

4.5.1.2.2.1 Impact. Any randomly selected painted or oxide coated portion of sample fixture shall be subjected to the impact test in the manner specified in 4.5.1.1.2.1. If necessary, in order to meet requirements of this test, portion to be tested may be separated from the fixture. Surface peeling shall be construed as test failure.

4.5.1.2.2.2 Adhesion (applicable to painted surfaces only). Any randomly selected painted portion of sample fixture shall be subjected to the adhesion test specified in 4.5.1.1.2.2. Flaking of finish shall be construed as test failure.

4.5.1.2.2.3 Hardness (applicable to painted surfaces only). Any randomly selected painted portion of sample fixture shall be subjected to the hardness test specified in 4.5.1.1.2.3. Cutting or scratching of finish shall be construed as test failure.

4.5.1.3 Thickness of enclosure and reflector. The over-all thickness of the bare(unfinished) enclosure and reflector shall be measured at five randomally selected points with a micrometer caliper accurate to plus or minus 0.001 (±.001) inch. No measurement shall be less than the thickness specified in 3.3.5.

4.5.2 End item testing.

4.5.2.1 Reflector finish.

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4.5.2.1.1 Thickness of finish (applicable to Class I fixture only). The thickness of Class I, porcelain enamel, finishes shall be measured in accordance with method 6181 of Fed. Test Method Std. No. 141 (see 3.3.8.1.1).

4.5.2.1.2 Average reflectance. Average reflectance shall be determined by averaging the reflectance values taken at five randomly selected points on reflector surface. The average value so obtained shall be at least 96 percent of the value given in independent test lab data (see 3.3.8.4 and 4.6.5). Reflectance measurements shall be made with a reflectometer or instrument providing equivalent results.

4.5.2.1.3 Uniformity of finish. Visually examine the reflector finish to assure that the enamel or aluminum oxide covers the reflector surface completely and uniformly. If, after visual examination, the uniformity of finish is suspect, measure five or more areas of the reflector, including areas of suspect uniformity, with a reflectometer or other instrument which will give equivalent results. A ten percent difference in any two measurements shall be cause for test failure.

4.5.2.2 Rigidity test. The complete fixture, with lamps inserted, shall be placed face down on a horizontal support surface. The fixture shall then be lifted by application of a vertical force at one corner until such time as lift corner has been raised at least one foot and three of the four fixture corners are clear of the horizontal support surface; the fourth corner (diagonally opposite life corner) only remaining on the support surface. When in correct test position, the fixture shall be supported only at the lift corner and the corner diagonally opposite lift corner. This position shall be maintained for at least one minute. Fixture and/or lamp damage or disassembly at any time during performance of test shall constitute test failure (see 3.3.12). The above test shall be repeated until fixture has been tested by lifting at each of four corners. Four samples per lot shall be tested. Failure of any fixture shall be basis for lot rejection.

4.5.3 Preproduction sample testing. Preproduction sample tests (see 4.2.1 and 3.3.16) shall include all tests in 4.6, the above tests cited in 4.5.1 and 4.5.2, and the following additional tests.

4.5.3.1 Reflector finish.

4.5.3.1.1 Selection and preparation of test panels. Class I test panels of the same material, thickness, and finish as specified for the Class I reflector in 3.3.5 and 3.3.8.1.1, and Class II and III test panels of the same material, thickness and finish as specified in 4.5.1.1.2 shall be prepared in accordance with Method 2011 of Fed. Test Method Std. No. 141. Following preparation panels shall be subjected to the tests specified in 4.5.3.1.1.1 thru and including 4.5.3.1.1.4.

4.5.3.1.1.1 Water corrosion. One test panel shall be subjected to the test of method 6061 of Fed. Test Method Std. No. 141 for 500 hours at 100 degrees F. (37.7 degrees C.) except that distilled water shall be substituted for the salt solution. At the end of test there shall be no evidence of corrosion. Evidence of corrosion shall be construed as test failure.

4.5.3.1.1.2 Salt-spray corrosion. One test panel shall have two diagonal cuts made through finish to the base metal and shall then be subjected for 250 hours to the salt-spray test of method 6061 of Fed. Test Method Std. No. 141. At the end of test there shall be no evidence of corrosion at distances greater than 3/16 inch from the cuts. Evidence of corrosion at distances greater than 3/16 inch shall be construed as test failure.

4.5.3.1.1.3 Moist-hydrogen-sulfide. Determine the average reflectance value of a test panel, by use of a reflectometer or other instrument which will give equivalent results, by measuring five or more randomly selected points on the panel and averaging the measured values. Then subject test panel to an atmosphere of hydrogen sulfide over water in a bell jar at 70 degrees F. (21° C) for 48 hours. Again, determine the average reflectance value in the manner outlined above. Compare the average reflectances as determined before and after the hydrogen-sulfide exposure. The loss in average reflectance shall not exceed 5 percent. Lost of more than five percent shall be construed as test failure.

4.5.3.1.1.4 Heat test. Determine the average reflectance of a test panel in the manner cited in 4.5.3.1.1.3. Then hang test panel in an electric oven at a temperature of 300 degrees F. (149 degrees C.) for 4 hours. Remove panel from oven and again determine the average reflectance in the above manner. Compare average reflectances as determined before and after placement in oven. Loss in average reflectance shall not exceed one percent. Loss of more than one percent shall be construed as test failure.

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4.5.3.1.2 Exposure test. Subject at least five sample plates, each flat, 4 inches wide by 6 inches long and representative of the reflector finish and material (see 3.3.5, 3.3.8.1 and 4.5.1.1.2) to testing in the following manner. Determine initial average reflectance of sample plates by measuring the reflectance of each sample with a reflectometer, or instrument providing equivalent results, and averaging the cumulative reflectance measurements. Then expose sample plates for one hundred hours to the radiation of a glass enclosed carbon arc lamp, such as a Fade-O-Meter, operating on a current of 13 + .5 amperes and electrical potential, at the arc, of 140 volts. Each sample shall be placed ten inches from the arc and the lamp so ventilated that the temperature at the sample soes not exceed 105° F. After exposure in the above manner, again determine the average reflectance as outlined above. Compare the initial average reflectance to average reflectance after exposure and ascertain compliance with the decrease of no more than 4 percent requirement of 3.3.8.4. Also, compare average reflectance value of test samples after exposure with average fixture reflectance value given in photometric test data (see 3.3.8.4 and 4.6.5). The exposure test need only be conducted for preproduction testing when a preproduction sample(s) is required (see 3.3.16), or only for the first lot of fixtures of each type, style, class, kind and size specified in the invitation for bids, when a preproduction sample(s) is not required. Inability to meet applicable reflectance requirements of 3.3.8.4 shall be cause for lot rejection.

4.5.3.2 Salt spray test. One complete sample fixture, less the electrical components, shall be prepared in accordance with method 2011 of Fed. Test Method Std. No. 141, whereupon it shall be subjected to the salt spray test specified in Method 811.1 of Fed. Test Method No. 151 for a period of 48 hours. Upon completion of test there shall be no evidence of corrosion on any portion of the fixture, including the fixture reflector. Evidence of corrosion shall be construed as test failure. See 3.3.8.1.4 and 3.3.9.1.

4.6 Photometric tests. All tests described in 4.6 shall be conducted on one complete fixture of each type, style, class, kind and size (see 1.2) specified in the Invitation for Bids. Additionally all tests must be conducted by an independent laboratory acceptable to the Government.

4.6.1 Shielding angle determination. Shielding angle shall be measured in a vertical plane, perpendicular to the fixture's longitudinional axis, with fixture orientated in the position of normal use. The minimum shielding angle shall be determined by measurement of the shielding angle in each of five randomly selected vertical planes. The smallest of the five angles so measured shall be the minimum shielding angle of the fixture and shall not be less than the angle specified in 3.3.15.1.

4.6.2 Determination of spacing-to-mounting height ratio. Spacing-to-mounting height ratio shall be determined in accordance with procedures outlined in Fig. 9-14, on page 9-34, of the IES Lighting Handbook, Fifth Edition (1972). See 3.3.15.2.

4.6.3 Determination of light distribution. In accordance with the procedures outlined on pages 4-16, 4-17 and 4-18 of the IES Light Handbook, Fifth Edition (1972) under the general section titled "Photometry of Luminaires" and subsections titled "General Lighting Luminaires (Candlepower Distribution and Luminance (Photometric Brightness));" "General Lighting Luminaires (Total Lumen Output)" and "Calculations of Luminaire Lumens From Candlepower Data," perform the following:

(a) Obtain fixture candlepower distribution data in at least 3 vertical planes; a vertical plane parallel to the fixture's lamps, a vertical plane perpendicular to the fixture's lamps and an intermediate vertical plane of 45 degrees; the vertical planes being with respect to the lamp longitudional axis. Candlepower readings shall be taken at intervals of 10 degrees or less in the center of each interval (zone).

(b) From the above candlepower data calculate the luminous flux values for each zone from nadir to 180 degrees above nadir.

(c) Compute the percentage of light distribution utilizing the luminous flux values calculated above and the following relations:

$$\text{Percent Upward Light Distribution} = \frac{\text{Lumen Output } (90^\circ \text{ to } 180^\circ \text{ zones}) \times 100}{\text{Lumen Output } (0^\circ \text{ to } 180^\circ \text{ zones})}$$

$$\text{Percent Downward Light Distribution} = \frac{\text{Lumen Output } (0^\circ \text{ to } 90^\circ \text{ zones}) \times 100}{\text{Lumen Output } (0^\circ \text{ to } 180^\circ \text{ zones})}$$

4.6.4 Efficiency. Fixture efficiency shall be determined in accordance with the procedures outlined on page 4-18 of the above mentioned IES Handbook under the section titled "General Lighting Luminaires (Total Lumen Output)."

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4.6.5 Photometric tests and test data. Any changes made to the fixture would in any way affect any requirement of 3.3.15 shall require the performance of new photometric tests in accordance with 4.6. Also, data furnished in compliance with 3.3.15 shall contain fixture reflectance value, and shall be no older than 5 years of age. Date (month and year) of testing shall be indicated on all test reports. Fixture reflectance shall be an average reflectance value, determined as specified in 4.5.2.1.2.

4.7 Inspection of preparation for delivery. An inspection shall be made to determine that packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table III. For examination of interior packaging, the sample unit shall be one shipping container prepared for delivery and selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

Table III. Classification of preparation for delivery

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect, illegible; improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of flaps, loose strapping, inadequate stapling. Distortion of the container.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.1).

5.1.1 Level A. Each fixture with mounting accessories shall be packaged in a close-fitting box conforming to PPP-B-636, class weather-resistant. Fiberboard blocking and bracing shall be provided with the fiberboard meeting the requirements of PPP-F-320, type CF, class domestic, variety SW, grade 200. The box shall be closed and waterproofed in accordance with the appendix to the box specification.

5.1.2 Level B. Each fixture with mounting accessories shall be packaged in a close-fitting box conforming to PPP-B-636, class domestic. Fiberboard blocking and bracing shall be provided with the fiberboard meeting the requirements of PPP-F-320, Type CF, class domestic, variety SW, grade 200. The box shall be closed in accordance with method 1 or 2 of the appendix to the box specification.

5.1.3 Level C. The fixture shall be packaged to afford adequate protection against corrosion and damage during shipment from the supplier to the initial destination. This level may conform to the suppliers commercial practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Unit packages of fixtures of like description in quantities as specified (see 6.1), shall be packed in a close-fitting box conforming to PPP-B-585, Class 3; PPP-B-591 Class II; PPP-B-601, overseas type; PPP-B-621, Class 2; or PPP-B-640, Class 2. The box shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight shall not exceed 200 pounds.

5.2.2 Level B. Unit packages of fixtures of like description in quantities as specified (see 6.1), shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with method II of the appendix to the box specification. The gross weight shall not exceed the weight limitations of the box specification. When specified (see 6.1), the box shall be class weather-resistant and closed method V.

5.2.3 Level C. The fixtures shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

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5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.1).

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.
(b) Type, style, class, kind and size fixture desired (see 1.2).
(c) Mounting means and mounting height and ceiling height (see 3.3.6).
(d) Finish, if other than specified (see 3.3.9).
(e) If flexible cord is desired and length of same (see 3.3.10).
(f) Electrical requirements (see 3.3.11).
(g) Marking, if other than specified (see 3.3.13).
(h) Preproduction sample, if not required (see 3.3.16).
(i) Responsibility for inspection, if other than specified (see 4.1).
(j) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
(k) Quantity of unit packages of fixtures required in shipping container (see 5.2.1 and 5.2.2)
(l) When PPP-B-636, Class weather-resistant box is required (see 5.2.2).
(m) Selection of Civil or Military marking requirements (see 5.3).

6.2 Other fixture types. Refer to the latest version of specification W-F-414 for coverage of pendant mounted office fixtures and to the latest version of W-F-1662 for coverage of recessed and surface ceiling office fixtures.

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